

The Impact of Commercial and Utility-Scale Solar Energy on High Value Farmland

Abstract

Purpose: The expected rapid development of solar energy in the United States has raised concerns about its effect on high valued farmland. This study examines how commercial and utility-scale solar energy facilities influence farmland prices across Indiana from 2015 to 2020.

Design/methodology: We estimate a hedonic price model to capture the relationship between farmland values and distance to the nearest solar facility. Quantile and expectile regressions are employed to measure effects across the conditional price distribution.

Findings: Our results provide consistent evidence that farmland closer to solar facilities sells at a premium. Traditional OLS results suggest a 1.4% price premium per mile closer to a solar facility, while the premium increases to 1.8% at the upper 90th expectile.

Implications: Our findings support the concern that solar energy development may impact high value farmland. These findings have significant implications for policymakers, highlighting the need to balance the economic benefits of solar energy expansion with the preservation of high value agricultural lands, ensuring that farmers, tenants and landowners are not disproportionately affected by increasing land costs.

Originality/value: This paper provides new evidence on how solar energy development shapes farmland values across the price distribution, extending the standard hedonic framework through the integration of quantile and expectile regressions.

Keywords: Expectile, Farmland, Quantile, Renewable Energy, Solar Energy
JEL code: Q15, Q24, R14, R52

1 Introduction

Renewable energy sources like solar, wind, geothermal, and hydropower currently account for about 13% of energy demand in the United States (EIA, 2023). However, by 2050, the share of U.S. energy demand from renewable sources is projected to increase to up to 50% (Hand et al., 2012). According to the U.S. Energy Information Administration, among the various renewable energy sources, the projected growth in solar generation is the most pronounced from 3% in 2020 to 20% in 2050 (EIA, 2021). There is a growing concern among many, including farmers and policymakers, that much of the anticipated growth in solar energy generation places high quality farmland at risk because solar energy generation may offer compensation that is much higher than prevailing cash rental rates or agricultural production returns (Huffstutter et al., 2024; Maguire et al., 2024). Further, the potential compensation from solar energy generation may be capitalized into neighboring farmland prices. A number of recent studies have examined the degree to which solar energy generation influences farmland prices (Abashidze and Taylor, 2022; Lai et al., 2019), yet these studies principally examine the relationship through regression analysis at the conditional mean. Given the concern of solar energy adversely affecting high quality farmland, this study examines the degree to which solar energy generation affects farmland prices across their conditional distribution.

Farmland prices are determined by a wide variety of factors, such as urban sprawl (Delbecq et al., 2014; Kuethe et al., 2011), natural amenities (Borchers et al., 2014; Uematsu et al., 2013), climate conditions (Fezzi and Bateman, 2015), and government payments (Ifft et al., 2015). The hedonic price method relates observed farmland transaction prices to quantifiable measures of these potential price determinants (Zhang and Nickerson, 2015). Typically, hedonic price models are estimated through ordinary least squares (OLS) regression methods which capture the relationship between the variation in each factor and the conditional mean of the price distribution. This approach, however, is not able to identify the degree to which these relationships may vary between low and high value properties. Previous studies show that this form of heterogeneity can be incorporated in hedonic price models through quantile regression estimation (Zietz et al.,

2008a; Kuethe and Keeney, 2012; Zhang, 2016; Mueller and Loomis, 2014). Quantile regression is a generalization of median regression under which the estimated coefficients vary across the conditional price distribution.

This study examines the degree to which commercial and utility-scale solar farms impact the market price of nearby agricultural lands, accounting for the potential of varying impacts across the conditional price distribution. We estimate an hedonic pricing model using more than 38,000 farmland transactions across the state of Indiana from 2015 to 2020. In order to evaluate the degree to which the distance from commercial or utility-scale solar facilities impacts farmland prices of differing quality grades, we estimate our hedonic pricing model using both quantile and expectile regressions. While quantile regression is a generalization of the median regression that minimizes absolute residuals, expectile regression is a generalization of mean regression which minimizes squared residuals. As a result, expectile regression is more directly comparable to the *status quo* OLS regression.

We contribute to the understanding of the relationship between solar energy generation and farm real estate values in three ways. First, our results provide consistent evidence that farmland closer to existing commercial or utility-scale solar facilities sells at a premium. This finding confirms prior evidence of the relationship between solar energy and agricultural land values in Taiwan by Lai et al. (2019) but contrasts the estimation results of Abashidze and Taylor (2022), who examine the relationship in North Carolina. While Abashidze and Taylor (2022) find no direct spillover effects on agricultural land values in North Carolina, they suggest that solar farm construction may indirectly influence land values by signaling the land's potential for future solar development. In contrast, our study provides direct evidence of spillover effects, with the impact being more pronounced for higher-value farmland, which may be region-specific to Indiana. The solar price premium for farmland also conflicts with the negative price pressure for residential real estate markets in prior studies (Dröes and Koster, 2021; Maddison et al., 2022). Second, by including a simple square of the distance to the nearest commercial or utility-scale solar facility, we also find consistent evidence that the price premium persists up to a set distance, beyond which it diminishes. Third, through both quantile and expectile regression, we find consistent evidence that

the price premium associated with commercial and utility-scale solar facilities is higher for high-value farmland. For example, traditional OLS estimates suggest a 1.4% premium for an additional mile closer to neighboring solar farms (at the conditional mean), yet the premium increases to 1.8% for the upper 90th expectile. Further, the premium is indistinguishable from zero at the 99 percent confidence interval for the lowest value farmland (10th quantile or expectile). Our approach extends the Average Treatment Effect (ATE) framework by using quantile and expectile regression to assess the differential impacts of solar energy development on farmland values across the price distribution. Thus, our findings support the concern that solar energy development is particularly pronounced for high value farmland.

The findings of this study have significant implications for policymakers, particularly as solar energy development continues to grow. The price premium associated with proximity to solar facilities suggests that farmland closer to solar developments is valued higher, which may have different impacts across various stakeholders. For example, farmers and tenants can face increased rental costs as landowners adjust rental rates to reflect higher land values, potentially reducing the profitability from agricultural operations. Similarly, solar companies may face higher land acquisition costs when trying to secure land for future solar projects. In contrast, for landowners, the price premium represents a potential financial gain, as their land could be leased or sold at higher prices. Given these competing effects, policymakers should consider these dynamics when designing policies to balance the economic benefits of solar energy development with the preservation of high-value farmland and the financial well-being of farmers and tenants.

The remainder of this paper is organized as follows. Section 2 introduces the conceptual framework of our hedonic price model and our empirical estimation approach using both quantile and expectile regressions. Section 3 outlines the data of our hedonic price model, including the farmland transaction record, the location of solar generation facilities, and the additional control variables. Section 4 summarizes the empirical results of our hedonic price model. Section 5 presents a robustness check of our findings. Finally, Section 6 provides a concluding summary of key findings.

2 Conceptual Framework

As previously mentioned, farmland prices are determined by a complex set of factors, including both farm and non-farm characteristics of the parcel (Zhang and Nickerson, 2015). Farm characteristics include attributes that affect agricultural output, such as soil productivity and parcel size. Non-farm characteristics include locational and environmental attributes that are not associated with agricultural output, such as distance to urban centers or the presence of natural amenities, including forests or bodies of water. Following Rosen (1974), the market price of a differentiated good is determined by the contribution of each of its attributes or characteristics. This so-called hedonic price method builds on the consumption theory of Lancaster (1966), which posits that consumers derive utility from the combination of characteristics or attributes of a good. Thus, the hedonic price of farmland can be expressed:

$$p_i = f(A_1, \dots, A_J, B_1, \dots, B_K) \quad (1)$$

where p_i is the market price of farmland parcel i . The market price is a function of the parcel's J farm characteristics (A) and K non-farm characteristics (B).

Empirically, the hedonic price model is used to specify a regression equation that captures the relationship between observed markets price of a good and each of its characteristics. The regression equation can be expressed as:

$$\log(p_i) = x_i^T \beta + \epsilon_i \quad (2)$$

where $\log(p_i)$ represents the natural logarithm of the observed market price of farmland parcel i . The vector x_i contains the quantifiable farm and non-farm characteristics of parcel i , following A_j and B_k in (1). The control variables in x may also include time and locational fixed effects. The vector β represents the unknown parameters estimated through the regression, and the vector ϵ_i is the regression error. We are primarily interested in the conditional correlation between farmland market prices and distance to the nearest utility or commercial-scale solar facility. When (2) is

estimated through Ordinary Least Squares (OLS), the coefficient estimate $\hat{\beta}$ on the distance to the nearest solar facility represents the capitalization of solar generation returns on farmland prices at the conditional mean.

2.1 Empirical framework

Given the public concern that expanding solar energy development will adversely affect high value farmland, we are interested in the impact of solar farms on farm real estate values above the conditional mean. A number of prior studies demonstrate that the hedonic price model (1) can also be estimated using quantile regression to examine pricing relationship across the conditional distribution (Zietz et al., 2008a; Ebru and Eban, 2011; Zhang, 2016; Kueth and Keeney, 2012; Zhang et al., 2020). A quantile is the percentile value of the population distribution. For example, the 0.1 quantile is the value of $\log(p_i)$ below which 10% of farmland prices fall.

When (2) is estimated using OLS, the regression minimizes the squared errors with the assumption that the sum of the errors is zero. As developed by Koenker and Bassett (1978), quantile regression minimizes the regression errors with the assumption that the absolute errors equal zero at the quantile of interest. Thus, under quantile regression (2) can be expressed:

$$\log(p_i) = x_i^T \beta + \epsilon_i \text{ with } \mu_\tau(\epsilon_i) = 0 \quad (3)$$

where $\mu_\tau(\epsilon)$ is the quantile function for the quantile $\tau \in (0, 1)$. For example, when we estimate (3) at $\tau = 0.1$, the estimated coefficient of interest captures the impact of solar farms on farmland prices at or below 10% of the population.

Quantile regression uses linear programming methods to minimize the absolute deviation between observed and predicted farmland prices using the loss function:

$$\left\{ \sum_{i=1}^n \omega_\tau^q(\epsilon_i) |\log(p_i) - \widehat{\log(p_i)}| \right\}, \quad (4)$$

$$\text{where } \omega_{\tau}^q(\epsilon) = \begin{cases} (1 - \tau) & \text{if } \epsilon \leq 0 \\ \tau & \text{if } \epsilon > 0 \end{cases}$$

where w_{τ}^q is a weighting function equal to τ for positive deviations and $(1 - \tau)$ for negative deviations. Quantile regression is a generalization of median regression, as $\tau = 0.5$ is equal to the conditional median of the population. However, by minimizing the absolute deviation in (4) the estimation results at $\tau = 0.5$ may not be directly comparable to the OLS conditional mean estimates when the conditional distribution of farmland prices does not follow strict normality assumptions.

As a result, we also estimate the hedonic price function (1) using expectile regression methods developed by Newey and Powell (1987). Instead of minimizing absolute deviations through linear programming, expectile regression methods minimize squared deviations through quadratic optimization. The expectile regression loss function can be similarly expressed:

$$\left\{ \sum_{i=1}^n \omega_{\tau}^e(\epsilon_i) \left(\log(p_i) - \widehat{\log(p_i)} \right)^2 \right\}, \quad (5)$$

$$\text{where } \omega_{\tau}^e(\epsilon) = \begin{cases} (1 - \tau) & \text{if } \epsilon \leq 0 \\ \tau & \text{if } \epsilon > 0 \end{cases}$$

As Philipps (2022) argues, expectile regression methods have more limited adoption in applied economics because the interpretation of an expectile is not as straightforward as that of a quantile. An expectile is the mean of a weighted distribution where the weights are determined by the distance from the expectile of interest. Specifically, the expectile τ of a distribution is equivalent to the mean of the distribution when values above τ are $\frac{\tau}{1-\tau}$ more frequent than observed (Philipps, 2022). Thus, for high values of a $\tau \in (0, 1)$, the expectile will be influenced more by data on the upper side of the distribution.

Expectile regression offers a number of potential advantages over the more frequently used quantile regression. Both quantile and expectile regressions provide a more complete understanding of the distribution of farmland prices, are efficient when the prices are non-normally distributed, and do not require sub-setting the data. However, given that expectile regression is

estimated using quadratic optimization, its loss function is continuously differentiable. The estimated coefficients of expectile regression therefore represent average treatment effects that considers weighted values of all observations (Philipps, 2022). Quantile regression coefficients, on the other hand, yield only point estimates of the treatment effect at a given quantile. When (5) is estimated at $\tau = 0.5$, the expectile regression results converge to OLS. Thus, expectile regression is a generalization of conditional mean regression, including OLS.

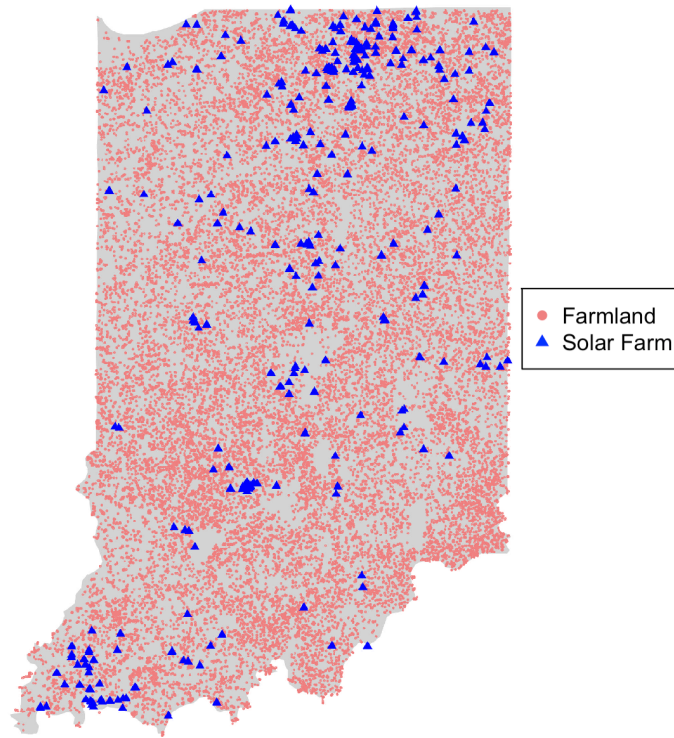
3 Data

We estimate the relationship between solar energy generation and farmland prices in Indiana from 2015 to 2020. Farmland prices were obtained from the transactions records maintained by the Indiana Land Sales Bulletin (ILSB). ILSB compiles farmland sales for transactions of 20 or greater acres filed at all 92 Indiana county courthouses. The transaction record contains information on parcel location, transaction date, sales price, acreage, and land use, including improvements (presence of buildings and structures). We remove all parcels with wetland and forests so that our analysis only considers the sale of agricultural land. The resulting transaction record includes 38,883 observed sales.

The relationship between each transaction and solar energy generation is captured by the distance to the nearest commercial or utility-scale solar facility, obtained from the Southern Indiana Renewable Energy Network (SIREN). SIREN is a mutual aid group formed to promote alternative energy. SIREN maintains a registry of the location, year of installation, wattage, type, and utility of solar generation facilities across Indiana. We limit our analysis to commercial and utility-scale solar facilities (types) because of the concern of increased land use for large scale solar development. The impact of these solar facilities on subsequent farmland transactions is measured by the Haversine or “as-the-crow-flies” distance between each transaction and the nearest existing commercial or utility-scale solar facility. Our study area, including observed transactions and commercial and utility-scale solar facilities, is depicted in Figure 1.

Our hedonic price model captures the relationship between distance to the nearest commercial

Figure 1: Study Area



or utility-scale solar facility and the observed price per acre obtained from the transaction record. We estimate the hedonic price model using a semi-log functional form in which the price per acre is expressed in natural logs and the control variables are expressed in levels. Thus, the estimated coefficients, when multiplied by 100, measure the approximate percentage change in per acre sales price for a unit change in the control variable.

In addition to the distance to the nearest solar facility, our hedonic model also includes a number of characteristics that may impact farmland transaction prices. The additional controls include farm characteristics, including total transacted acres obtained from ILSB and soil productivity ob-

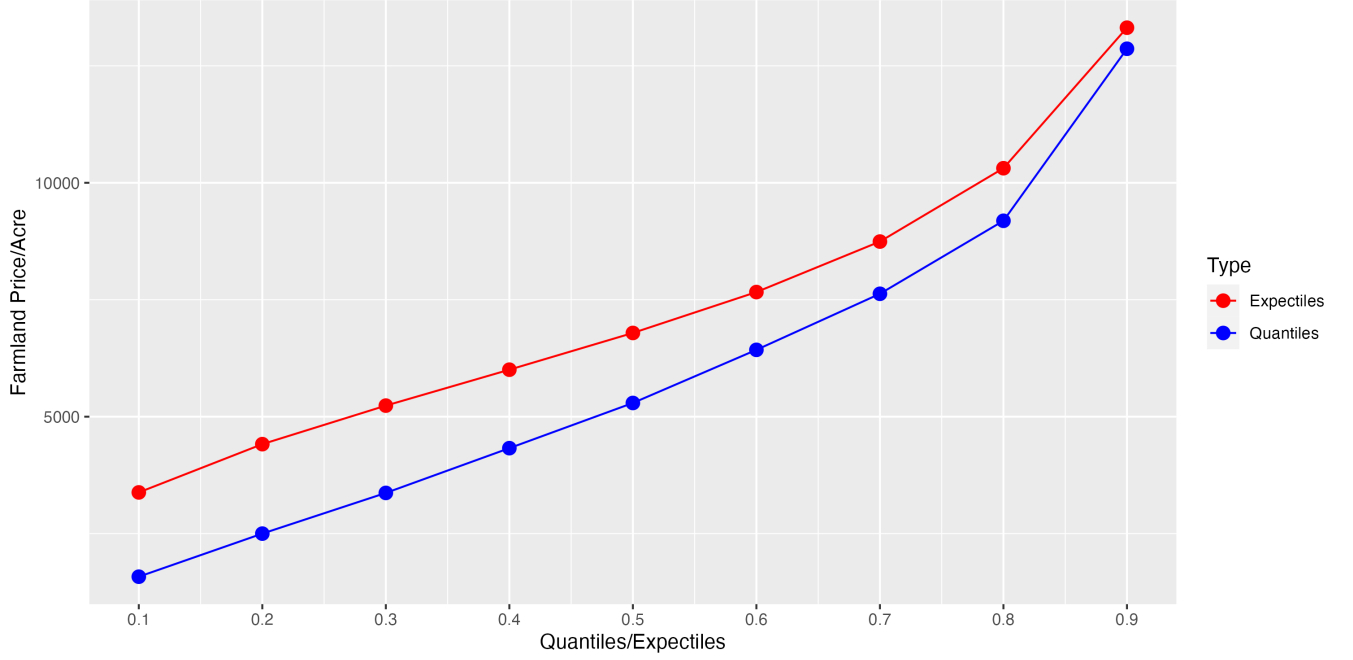
tained from the National Commodity Crop Productivity Index (NCCPI). NCCPI is calculated by the USDA National Resource Conservation Service and ranges from 0 to 1, with 1 as the highest productivity soils. The non-farm characteristics include several measures obtained from the U.S. Census Bureau, including distance to major cities (Chicago and Indianapolis), nearest town of at least 50,000 residents, and county population density (people per square mile). In addition, we control for the presence of electric transmission lines which are associated with a positive price premium for farmland (Lu et al., 2023) and may be correlated with solar development, obtained from the Indiana Geographic Information Office (indianamap.org). Finally, our hedonic price model also includes county and year fixed effects to control for unmeasured regional variation and common price movements across time. Table 1 reports the summary statistics for the variables considered in our hedonic price model.

Table 1: Summary Statistics for Transaction-Level Farmland Sales, 2015–2020

Variable	Mean	Std. Dev.	Min	Max
log (Price/Acre)	8.46	0.92	4.59	10.82
Distance to solar energy (miles)	15.31	11.46	0.05	73.64
Acres	82.49	96.41	20	1564
NCCPI (index, 0-1)	0.48	0.22	0	0.96
Distance to nearest city (miles)	72.47	30.91	1.004	168.07
Distance to nearest town (miles)	26.44	14.77	0.29	83.33
Population density (people per square miles)	188.34	291.5	21.27	2439.86
Distance to nearest PML (miles)	14.42	12.23	0.008	60.54

Given the concern of solar energy adversely affecting high quality farmland, we also estimate our hedonic price model using both quantile and expectile regression. Figure 2 plots the unconditional quantile and expectile across the observed per acre farmland price distribution. For high value farmland prices, we observed that the upper ten percent of values exceed \$12,865.57 per acre (the 0.9 quantile value). The conditional mean when these upper values are weighted by $(0.9/(1 - 0.9))$ is \$13,315.64 (the 0.9 expectile).

Figure 2: Distribution of land value per acre



4 Results

Our baseline estimation of the relationship between solar energy generation and farmland prices in Indiana through OLS is reported in the first column of Table 2. The estimated coefficient on the distance to the nearest commercial or utility-scale solar facility is statistically significant at the 99% confidence interval and negative. Specifically, the estimated coefficient implies that, at the conditional mean, the per acre price of farmland is expected to decline by 1.4% for each additional mile from the nearest commercial or utility-scale solar generation facility. Given that this price premium may decline at greater distances, we also include the squared distance, which is statistically significant and positive but small. Thus, the price premium associated with nearby solar generation facilities is expected to decline at a modest rate.

Table 2: Quantile Regression Result

	Dependent variable:									
	(OLS)	(0.1)	(0.2)	(0.3)	(0.4)	(0.5)	(0.6)	(0.7)	(0.8)	(0.9)
Solar_Distance	-0.014*** (0.003)	-0.008 (0.005)	-0.009*** (0.003)	-0.012*** (0.002)	-0.012*** (0.001)	-0.014*** (0.001)	-0.012*** (0.001)	-0.015*** (0.002)	-0.016*** (0.002)	-0.012*** (0.003)
(Solar_Distance) ²	0.0001** (0.00008)	-0.00005 (0.0001)	0.00002 (0.00009)	0.0001** (0.00005)	0.0001** (0.00005)	0.0002*** (0.00004)	0.0001*** (0.00004)	0.0002*** (0.00004)	0.0003*** (0.00005)	0.0002*** (0.00007)
Acres	-0.001*** (0.00009)	-0.001*** (0.0001)	-0.001*** (0.00008)	-0.001*** (0.00005)	-0.001*** (0.00004)	-0.001*** (0.00004)	-0.001*** (0.00003)	-0.001*** (0.00003)	-0.001*** (0.00005)	-0.001*** (0.00006)
NCCPI	0.15*** (0.047)	0.18*** (0.061)	0.25*** (0.041)	0.30*** (0.026)	0.27*** (0.023)	0.23*** (0.021)	0.17*** (0.019)	0.11*** (0.020)	0.038 (0.025)	0.005 (0.032)
Nearest City	-0.002* (0.0016)	-0.006*** (0.0020)	-0.007*** (0.0013)	-0.004*** (0.0008)	-0.002*** (0.0007)	-0.002*** (0.0008)	-0.002*** (0.0006)	-0.002*** (0.0006)	-0.003*** (0.0008)	-0.003*** (0.0010)
Nearest Town	-0.0005 (0.001)	0.009*** (0.002)	0.007*** (0.001)	0.0008 (0.001)	-0.0008 (0.0009)	-0.0024*** (0.0008)	-0.004*** (0.0008)	-0.006*** (0.0008)	-0.007*** (0.0009)	-0.007*** (0.001)
Density	0.001 (0.002)	-0.001 (0.0022)	-0.0015 (0.0007)	-0.0005 (0.0009)	0.0007 (0.0008)	0.002*** (0.0007)	0.004*** (0.0003)	0.003*** (0.0007)	0.006*** (0.0009)	0.005*** (0.0011)
Power Transmission Line	-0.002*** (0.0008)	-0.004*** (0.001)	-0.002*** (0.0007)	-0.001*** (0.0004)	-0.001*** (0.0004)	-0.0008** (0.0003)	-0.0006** (0.0003)	-0.0008** (0.0003)	-0.001*** (0.0004)	-0.0008 (0.0005)
Constant	8.43*** (0.008)	8.92*** (0.316)	9.35*** (0.212)	9.29*** (0.135)	9.30*** (0.119)	9.27*** (0.110)	9.24*** (0.100)	9.41*** (0.101)	9.53*** (0.130)	9.81*** (0.167)
Observations	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883
Pseudo R^2	0.181	0.086	0.102	0.125	0.137	0.141	0.138	0.130	0.128	0.150

Note: Standard errors in parenthesis.
Year and County fixed effects coefficients are not reported in the table.
*p<0.1; **p<0.05; ***p<0.01

As previously stated, farmers and policymakers have expressed concern that future solar development may adversely impact high quality farmland. Following a number of previous studies that also address price impacts along the conditional price distribution, we also estimate our hedonic price model using quantile regression (Zietz et al., 2008b; Heintzelman, 2010; Kuethe and Keeney, 2012; Uematsu et al., 2013). The remaining columns of Table 2 report the quantile regression results for quantiles 0.1 through 0.9. The quantile regression results provide two important findings about the relationship between solar energy facilities and farmland prices. First, the estimated coefficient for distance to the nearest solar facility is statistically indistinguishable from zero at the lowest quantile (0.1). Thus, solar generation facilities likely have minimal impact on low value farmland. Second, the estimated impact of solar energy facilities is more pronounced at higher quantiles than at lower quantiles. For example, the per acre price of farmland is expected to decline by 0.9% at the 0.2 quantile but by 1.2% for the 0.9 quantile. Thus, the estimated coefficients are consistent with the concerns expressed by farmers and policymakers. In addition, the estimated coefficients on the squared distance measure suggest that the price premium declines at greater distances for quantiles at or above 0.3. For both the OLS and quantile regressions, the estimated coefficients on the remaining control variables generally confer to prior expectations.

The quantile regression coefficients yield point estimates of the treatment effect at a given quantile, that minimize absolute deviations. Thus, the OLS and quantile regressions may not be directly comparable if conditional farmland prices are not distributed log-normally. We therefore also estimate our hedonic price function through expectile regression. Expectile regression coefficients yield estimates of the average treatment effect that considers weighted values of all observations using a quadratic loss function, with a greater weight placed on observations near the expectile of interest (Philipps, 2022). In contrast to quantile regression, expectile regression includes OLS estimates when the expectile of interest is 0.5. Table 3 reports the expectile regression results for our hedonic price model. Similar to Table 2, we report the coefficient estimates for expectiles 0.1 through 0.9, but because expectile regression considers the weighted value of all observations, we also include the extreme low and high prices at expectiles 0.01 and 0.99, respectively.

Using expectile regression, the estimated coefficients on the distance to the nearest commercial

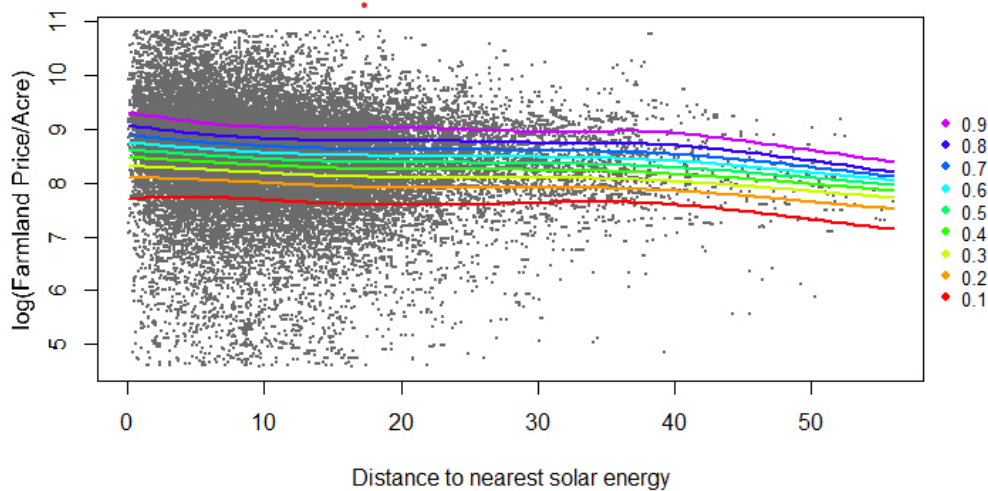
Table 3: Expectile Regression Result:

Dependent variable:											
	log(Price / Acre)										
	(0.01)	(0.1)	(0.2)	(0.3)	(0.4)	(0.5)	(0.6)	(0.7)	(0.8)	(0.9)	(0.99)
Solar_Distance	-0.003 (0.009)	-0.008* (0.003)	-0.011*** (0.003)	-0.012*** (0.003)	-0.013*** (0.003)	-0.014*** (0.003)	-0.015*** (0.003)	-0.016*** (0.003)	-0.017*** (0.003)	-0.018*** (0.003)	-0.021** (0.008)
(Solar_Distance) ²	0.00002 (0.0002)	0.00003 (0.00008)	0.00008 (0.00008)	0.00009 (0.00008)	0.0001* (0.00008)	0.0001** (0.00008)	0.0002** (0.00008)	0.0002*** (0.00008)	0.0002*** (0.00008)	0.0002*** (0.00008)	0.0003*** (0.0002)
Acres	-0.001*** (0.0002)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.00009)	-0.001*** (0.0002)
NCCPI	0.041 (0.102)	0.15*** (0.047)	0.15*** (0.047)	0.16*** (0.047)	0.16*** (0.047)	0.15*** (0.047)	0.13*** (0.047)	0.10*** (0.047)	0.07 (0.047)	0.025 (0.047)	-0.049 (0.102)
Nearest City	0.001 (0.003)	-0.002 (0.0016)	-0.003** (0.0016)	-0.003* (0.0016)	-0.002* (0.0016)	-0.002* (0.0016)	-0.002* (0.0016)	-0.002** (0.0016)	-0.003* (0.0016)	-0.003** (0.0016)	-0.005 (0.003)
Nearest Town	0.003 (0.003)	0.004* (0.001)	0.003* (0.002)	0.001 (0.001)	0.00006 (0.001)	-0.0005 (0.001)	-0.0001* (0.001)	-0.002* (0.001)	-0.004** (0.001)	-0.005*** (0.001)	-0.007*** (0.003)
Density	-0.0006 (0.003)	-0.0001 (0.0008)	0.0003 (0.002)	0.0006 (0.002)	0.0011 (0.002)	0.0016 (0.002)	0.0021 (0.002)	0.0027 (0.002)	0.0031 (0.002)	0.0037** (0.002)	0.0018 (0.003)
Power Transmission Line	-0.0028* (0.001)	-0.003*** (0.0008)	-0.002*** (0.0008)	-0.002*** (0.008)	-0.002*** (0.008)	-0.002*** (0.0008)	-0.002*** (0.0008)	-0.002*** (0.0008)	-0.002*** (0.0008)	-0.002*** (0.0008)	-0.002 (0.0027)
Constant	6.541*** (0.019)	7.635*** (0.008)	7.963*** (0.008)	8.158*** (0.008)	8.30*** (0.008)	8.43*** (0.008)	8.55*** (0.008)	8.67*** (0.008)	8.82*** (0.008)	9.04*** (0.008)	9.63*** (0.019)
Observations	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883
R ²	0.38	0.26	0.22	0.20	0.19	0.19	0.18	0.18	0.188	0.18	0.19
Note:	Standard errors in parenthesis. Year and County fixed effects coefficients are not reported in the table. *p<0.1; **p<0.05; ***p<0.01										

Note: Standard errors in parenthesis.
Year and County fixed effects coefficients are not reported in the table.
*p<0.1; **p<0.05; ***p<0.01

or utility-scale solar facility are statistically significant at the 99% confidence interval and negative for expectiles between 0.2 and 0.9. However, it should be noted that the relationship is weakly significant for the 0.1 expectile (90% confidence interval) and the 0.99 expectile (95% confidence interval). Similar to the quantile regression estimates, the estimated impact of solar energy facilities is more pronounced at higher expectiles than at lower expectiles. The estimated coefficients imply that the per acre price of farmland is expected to decline by 1.1% for each additional mile at the 0.2 expectile but by 1.8% at the 0.9 expectile. For farmland prices near the conditional mean (0.4 – 0.6 expectiles), the squared distance is statistically significant at the 90% to 95% significance, and for higher valued farmland (0.7 – 0.99 expectiles), the squared distance is statistically significant at the 99% confidence interval. Thus, the average treatment effect is expected to decline as the distance to the nearest solar facility increases beyond some threshold. This curvilinear relationship is directly observable in Figure 3. Thus, our expectile regression estimates provide additional evidence in support of the concerns that solar development may impact high value farmland.

Figure 3: Relationship between price per acre and distance to solar energy at observed expectiles



5 Robustness Check

To ensure the robustness of our findings, we conduct a series of additional analysis using alternative model specifications and sample restrictions. The results, presented in the table in the Appendix, confirm the consistency and reliability of our main findings.

First, to address potential concerns that our results might be driven by extremely large farmland transactions, we reestimated the quantile regression model by capping the price per acre at \$30,000. This adjustment ensures that our estimates are not disproportionately influenced by outliers. As shown in A1, the results remain consistent with our main findings, indicating that the price premium associated with proximity to solar facilities persists even when controlling for large transactions. The coefficients for distance to solar facilities remain statistically significant and negative across most quantiles, with the premium being more pronounced for higher-value farmland.

Second, we estimate our models without county fixed effects to evaluate the potential influence of regional variations. The results in A2 show that the coefficients for the distance to the solar facilities remain negative and statistically significant, although the magnitude of the coefficients increases slightly. This suggests that while county-level heterogeneity plays a role in farmland prices, the relationship between solar facilities and farmland prices is robust to the exclusion of county-fixed effects.

Third, we estimate our models excluding distance from power transmission lines (A3) and population density (A4) to examine whether the exclusion of these control variables influence our primary results. In both cases, we find that the coefficients for distance to solar facilities remain negative and statistically significant, with the premium being more pronounced for higher value farmland. This suggests that the relationship between solar facilities and farmland prices is robust to the exclusion of these control variables.

Furthermore, we conduct a quantile regression analysis at the extreme ends of the distribution (0.01 and 0.99 quantiles) as shown in A5. The results show that the price premium associated with solar facilities is not statistically significant for the lowest value farmland (0.01 quantile),

but it remains significant for the highest value farmland (0.99 quantile). This further supports our main finding that the price premium is more pronounced for high-value farmland. Finally, we segregate the data to focus on the upper 20% of farmland sales and apply an OLS regression model to that subset (A6). The results confirm that the price premium associated with proximity to solar facilities is particularly strong for high-value farmland. The coefficient for distance to solar facilities remains negative and statistically significant, with a magnitude similar to that observed in the quantile regression analysis.

These robustness checks confirm that our main findings are consistent across different model specifications and subsets of the data. The price premium associated with proximity to solar facilities is robust to the exclusion of key control variables, the inclusion of extreme quantiles, and the capping of large transactions. These results provide strong evidence that the impact of solar energy development on farmland prices is particularly pronounced for high-value farmland.

6 Conclusion

There is a growing concern that the anticipated growth in solar energy generation may adversely affect high valued farmland. The expected returns from solar energy generation exceed the returns of agricultural production and may be capitalized into neighboring farmland prices. We examine farmland transaction across Indiana from 2015 to 2020 to estimate the impact of the proximity to commercial and utility-scale solar facilities. Following previous studies, we estimate a hedonic price model using both ordinary least squares (OLS), which captures the relationship at the conditional mean, and quantile regression, which yields point estimates of the relationship across the conditional price distribution. In addition, we estimate our hedonic price model using expectile regression, which yields estimates of the average weighted treatment effect.

Our estimation results suggest that farmland closer to existing commercial or utility-scale solar facilities sells at a premium. The quantile and expectile regression results suggest that this premium is particularly pronounced for high value farmland. Further, this premium persists up to a set distance, beyond which it declines. Finally, our analysis suggests that solar generation facilities

have little price impact on the lowest value farmland.

Given the concerns expressed by farmers and policymakers, these findings may be useful in developing future policies or management advice. First, our study provides consistent evidence that the anticipated growth in solar energy development will likely impact the price of neighboring farmland. Second, the price impacts are higher for high value farmland. These premiums create competing interests, benefiting landowners through higher values while potentially increasing costs for farmers and solar developers. Thus, the anticipated growth in solar energy will likely influence the price and use of high value farmland. While the estimated price impacts are relatively modest, there is a concern that the potential conversion of high value farmland to energy production could alter the production and supply of agricultural commodities. Policymakers should consider these dynamics when designing policies to balance the economic benefits of solar energy development with the preservation of high-value farmland and the financial well-being of farmers and tenants. Our analysis suggests that continued research on the effect of solar energy generation on land use and agricultural production is warranted.

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Appendix

Table A1: Quantile Regression Results(Cap of Price per Acre at \$30000)

	Dependent variable:								
	log(Price/Acre)								
	Q0.1 (1)	Q0.2 (2)	Q0.3 (3)	Q0.4 (4)	Q0.5 (5)	Q0.6 (6)	Q0.7 (7)	Q0.8 (8)	Q0.9 (9)
Solar_Distance	-0.006** (0.003)	-0.009*** (0.002)	-0.010*** (0.001)	-0.010*** (0.002)	-0.011*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.008*** (0.001)
(Solar_Distance) ²	-0.0001 (0.0001)	0.00001 (0.00005)	0.0001* (0.00003)	0.0001* (0.00005)	0.0002*** (0.00005)	0.0002*** (0.00004)	0.0002*** (0.00002)	0.0002*** (0.00004)	0.0002*** (0.00005)
Acres	-0.001*** (0.0001)	-0.001*** (0.00004)	-0.001*** (0.00002)	-0.001*** (0.00004)	-0.001*** (0.0001)	-0.001*** (0.00002)	-0.001*** (0.00002)	-0.001*** (0.00003)	-0.001*** (0.00002)
NCCPI	0.208*** (0.026)	0.267*** (0.019)	0.320*** (0.014)	0.292*** (0.018)	0.250*** (0.015)	0.205*** (0.015)	0.139*** (0.012)	0.074*** (0.018)	0.023 (0.017)
Nearest City	-0.005*** (0.001)	-0.006*** (0.001)	-0.003*** (0.001)	-0.002*** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.001** (0.0004)	-0.002*** (0.001)	-0.004*** (0.0004)
Nearest Town	0.009*** (0.001)	0.008*** (0.001)	0.002*** (0.001)	0.0001 (0.001)	-0.002*** (0.001)	-0.004*** (0.001)	-0.005*** (0.0004)	-0.006*** (0.001)	-0.006*** (0.0005)
Density	-0.003 (0.003)	-0.003** (0.001)	-0.001*** (0.0004)	-0.001* (0.0005)	0.001** (0.0004)	0.002** (0.001)	0.002** (0.001)	0.003*** (0.001)	0.003*** (0.001)
Power Transmission Line	-0.004*** (0.001)	-0.002*** (0.0004)	-0.001*** (0.0002)	-0.001*** (0.0003)	-0.001*** (0.0002)	-0.001** (0.0003)	-0.001*** (0.0001)	-0.001*** (0.0002)	-0.001*** (0.0002)
Constant	8.931*** (0.414)	9.349*** (0.190)	9.274*** (0.081)	9.336*** (0.072)	9.275*** (0.056)	9.321*** (0.091)	9.476*** (0.110)	9.694*** (0.114)	9.978*** (0.110)
Observations	38,305	38,305	38,305	38,305	38,305	38,305	38,305	38,305	38,305

Note: Standard errors in parenthesis.
Year and County fixed effects coefficients are not reported in the table.
*p<0.1; ** p<0.05; ***p<0.01

Table A2: Quantile Regression Results(No County fixed effects)

	Dependent variable:								
	log(Price/Acre)								
	Q0.1 (1)	Q0.2 (2)	Q0.3 (3)	Q0.4 (4)	Q0.5 (5)	Q0.6 (6)	Q0.7 (7)	Q0.8 (8)	Q0.9 (9)
Solar_Distance	-0.015*** (0.004)	-0.019*** (0.002)	-0.022*** (0.002)	-0.018*** (0.002)	-0.018*** (0.002)	-0.019*** (0.002)	-0.017*** (0.001)	-0.013*** (0.002)	-0.016*** (0.002)
(Solar_Distance) ²	0.0004*** (0.0001)	0.0004*** (0.00005)	0.0004*** (0.00003)	0.0003*** (0.0001)	0.0003*** (0.00004)	0.0004*** (0.0001)	0.0004*** (0.00004)	0.0003*** (0.0001)	0.0004*** (0.00003)
Acres	-0.002*** (0.0003)	-0.002*** (0.0001)	-0.002*** (0.0001)	-0.001*** (0.00005)	-0.001*** (0.0001)	-0.001*** (0.00004)	-0.001*** (0.0001)	-0.001*** (0.00003)	-0.001*** (0.00003)
NCCPI	0.326*** (0.055)	0.459*** (0.029)	0.543*** (0.025)	0.530*** (0.023)	0.398*** (0.021)	0.291*** (0.020)	0.244*** (0.019)	0.126*** (0.023)	0.085*** (0.026)
Nearest City	-0.006*** (0.0005)	-0.006*** (0.0003)	-0.005*** (0.0003)	-0.004*** (0.0002)	-0.003*** (0.0002)	-0.003*** (0.0002)	-0.002*** (0.0001)	-0.001*** (0.0002)	-0.002*** (0.0002)
Nearest Town	-0.0004 (0.001)	-0.002*** (0.001)	-0.004*** (0.001)	-0.006*** (0.0004)	-0.007*** (0.0004)	-0.008*** (0.0004)	-0.008*** (0.0003)	-0.008*** (0.0005)	-0.006*** (0.0005)
Density	0.00002 (0.0001)	0.0001*** (0.00003)	0.0002*** (0.00003)	0.0003*** (0.00004)	0.0004*** (0.00001)	0.0004*** (0.00001)	0.0004*** (0.00002)	0.001*** (0.00003)	0.001*** (0.00003)
Power Transmission Line	-0.017*** (0.001)	-0.007*** (0.001)	-0.005*** (0.0005)	-0.004*** (0.0004)	-0.004*** (0.0004)	-0.004*** (0.0003)	-0.004*** (0.0003)	-0.004*** (0.0004)	-0.004*** (0.0004)
Constant	8.268*** (0.074)	8.433*** (0.038)	8.693*** (0.033)	8.855*** (0.031)	9.060*** (0.025)	9.238*** (0.024)	9.334*** (0.022)	9.504*** (0.030)	9.767*** (0.033)
Observations	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883

Note: Standard errors in parenthesis.
Year fixed effects coefficients are not reported in the table.
*p<0.1; **p<0.05; ***p<0.01

Table A3: Quantile Regression Results(Without Hub Distance)

	Dependent variable:								
	log(Price/Acre)								
	Q0.1 (1)	Q0.2 (2)	Q0.3 (3)	Q0.4 (4)	Q0.5 (5)	Q0.6 (6)	Q0.7 (7)	Q0.8 (8)	Q0.9 (9)
Solar_Distance	-0.007*** (0.002)	-0.010*** (0.002)	-0.012*** (0.001)	-0.013*** (0.001)	-0.015*** (0.002)	-0.013*** (0.001)	-0.015*** (0.001)	-0.017*** (0.001)	-0.014*** (0.001)
(Solar_Distance) ²	-0.0001 (0.0001)	0.00003 (0.00005)	0.0001*** (0.00003)	0.0001*** (0.00004)	0.0002*** (0.00005)	0.0002*** (0.00003)	0.0003*** (0.00002)	0.0003*** (0.00004)	0.0003*** (0.00004)
Acres	-0.001*** (0.0001)	-0.001*** (0.00004)	-0.001*** (0.00003)	-0.001*** (0.00003)	-0.001*** (0.0001)	-0.001*** (0.00001)	-0.001*** (0.00001)	-0.001*** (0.00002)	-0.001*** (0.00002)
NCCPI	0.220*** (0.017)	0.248*** (0.022)	0.303*** (0.013)	0.265*** (0.017)	0.232*** (0.016)	0.179*** (0.015)	0.112*** (0.012)	0.044** (0.018)	-0.017 (0.020)
Nearest City	-0.006*** (0.001)	-0.007*** (0.001)	-0.004*** (0.001)	-0.003*** (0.001)	-0.002*** (0.001)	-0.002*** (0.0004)	-0.002*** (0.0004)	-0.003*** (0.001)	-0.004*** (0.0005)
Nearest Town	0.010*** (0.001)	0.007*** (0.001)	0.001 (0.001)	-0.001 (0.001)	-0.003*** (0.001)	-0.005*** (0.001)	-0.006*** (0.0005)	-0.007*** (0.001)	-0.007*** (0.001)
Density	-0.001 (0.003)	-0.002* (0.001)	-0.0004 (0.001)	0.001 (0.0005)	0.002*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.006*** (0.001)	0.005*** (0.001)
Constant	8.735*** (0.343)	9.336*** (0.177)	9.317*** (0.100)	9.325*** (0.067)	9.267*** (0.099)	9.241*** (0.111)	9.361*** (0.131)	9.531*** (0.076)	9.871*** (0.083)
Observations	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883

Note: Standard errors in parenthesis.

Year and County fixed effects coefficients are not reported in the table.

* p<0.1; ** p<0.05; *** p<0.01

Table A4: Quantile Regression Results(Without Density)

	<i>Dependent variable:</i>								
	Q0.1 (1)	Q0.2 (2)	Q0.3 (3)	Q0.4 (4)	Q0.5 (5)	Q0.6 (6)	Q0.7 (7)	Q0.8 (8)	Q0.9 (9)
Solar_Distance	-0.008*** (0.003)	-0.010*** (0.002)	-0.013*** (0.001)	-0.012*** (0.002)	-0.014*** (0.002)	-0.013*** (0.001)	-0.015*** (0.001)	-0.017*** (0.001)	-0.012*** (0.001)
(Solar_Distance) ²	-0.0001 (0.0001)	0.00003 (0.00005)	0.0001*** (0.00003)	0.0001*** (0.00005)	0.0002*** (0.00005)	0.0002*** (0.00003)	0.0003*** (0.00003)	0.0003*** (0.00003)	0.0003*** (0.00004)
Acres	-0.001*** (0.0001)	-0.001*** (0.00004)	-0.001*** (0.00002)	-0.001*** (0.00004)	-0.001*** (0.00001)	-0.001*** (0.00002)	-0.001*** (0.00001)	-0.001*** (0.00001)	-0.001*** (0.00002)
NCCPI	0.182*** (0.028)	0.260*** (0.024)	0.307*** (0.011)	0.269*** (0.020)	0.236*** (0.017)	0.177*** (0.016)	0.117*** (0.012)	0.037** (0.017)	-0.006 (0.014)
Nearest City	-0.006*** (0.001)	-0.007*** (0.001)	-0.004*** (0.0005)	-0.003*** (0.001)	-0.002*** (0.001)	-0.002*** (0.0005)	-0.002*** (0.0005)	-0.003*** (0.001)	-0.004*** (0.0004)
Nearest Town	0.009*** (0.001)	0.008*** (0.001)	0.001 (0.001)	-0.001 (0.001)	-0.002*** (0.001)	-0.005*** (0.001)	-0.006*** (0.001)	-0.007*** (0.001)	-0.007*** (0.0005)
Power Transmission Line	-0.005*** (0.001)	-0.002*** (0.0004)	-0.001*** (0.0002)	-0.001*** (0.0004)	-0.001*** (0.0003)	-0.001** (0.0003)	-0.001*** (0.0002)	-0.001*** (0.0003)	-0.001*** (0.0002)
Constant	8.745*** (0.204)	9.194*** (0.161)	9.245*** (0.063)	9.373*** (0.065)	9.499*** (0.051)	9.606*** (0.049)	9.815*** (0.069)	10.159*** (0.053)	10.404*** (0.045)
Observations	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883	38,883

Note:

Standard errors in parenthesis.

Year and County fixed effects coefficients are not reported in the table.

* p<0.1; ** p<0.05; *** p<0.01

Table A5: Quantile Regression Results (0.01 and 0.99)

	<i>Dependent variable:</i>	
	log(Price/Acre)	
	Q0.01 (1)	Q0.99 (2)
Solar.Distance	−0.004 (0.005)	−0.011*** (0.002)
(Solar.Distance) ²	0.00005 (0.0002)	0.0001 (0.0001)
Acres	−0.001*** (0.0002)	−0.002*** (0.00005)
NCCPI	−0.075 (0.076)	−0.040*** (0.014)
Nearest City	−0.005*** (0.002)	−0.006*** (0.001)
Nearest Town	0.010*** (0.003)	−0.007*** (0.001)
Density	−0.001 (0.006)	0.0001 (0.001)
Power Transmission Line	−0.001 (0.001)	−0.004*** (0.0003)
Constant	7.142*** (0.601)	11.050*** (0.156)
Observations	38,883	38,883

Note:

Standard errors in parenthesis.

Year and County fixed effects coefficients are not reported in the table.

*p<0.1; **p<0.05; ***p<0.01

Table A6: OLS Regression Results (Top 20% Quantiles)

	<i>Dependent variable:</i>
	log(Price/Acre)
Solar_Distance	−0.013*** (0.002)
(Solar_Distance) ²	0.0003*** (0.0001)
Acres	−0.001*** (0.0001)
NCCPI	−0.143*** (0.024)
Nearest City	−0.005*** (0.001)
Nearest Town	−0.004*** (0.001)
Density	0.003*** (0.001)
Power Transmission Line	−0.002*** (0.0004)
Constant	9.985*** (0.095)
Observations	7,784
R ²	0.186
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	